

ICT1011: Computer Organisation and Architecture

Tutorial 2: Central Processing Unit

Q1. Computer Arithmetic and the Status Flags

For each of the following arithmetic operation, describe what result is obtained and which of the status flags (i.e. Negative, Zero, Overflow, Carry) are set:

- (Q1a) $10011001_2 + 11000000_2$ (binary)
- (Q1b) $10110101_2 + 11010001_2$ (binary)
- (Q1c) $H'FF + H'01$ (hexadecimal)
- (Q1d) $H'26 + H'5A$ (hexadecimal)
- (Q1e) $10101010_2 - 10101010_2$ (binary)
- (Q1f) $00110010_2 - 01001100_2$ (binary)
- (Q1g) $H'CF - H'89$ (hexadecimal)
- (Q1h) $H'83 - H'11$ (hexadecimal)

For the subtraction operations (Q1e – Q1f), you can assume that the CPU will perform a 2's complement addition and the status flags will set/cleared based on this addition operation.

Q2. CPU Internal Operations

Figure Q2 shows the internal connectivity of a simple CPU which may not be a MSP430. Buses, registers, memory, instruction register (IR), an ALU and program counter (PC) are shown (but not control logic). It also has a condition code register (CCR) with flags for Z, N, C, V. Arrows on the various buses show their direction. You don't need to know how 'wide' or how 'fast' this CPU is (it could be 8-bit, but equally well could be 64-bit).

We reset this CPU, so that all memory and registers begin with the value zero, and then insert the following instructions into memory locations 0 and 1 respectively:

MOV #6, R5 ; Move the value of 6 into R5

ADD R5, R6 ; R6 = R6 + R5

- Q2a) After running this program, what would you expect to be the content of registers R5 and R6?
- Q2b) If we start at the beginning and run through the execution of the first instruction, trace through the sequence of what happens using the table Q2. Concentrate on the order of events rather than the exact timing (there should be about 5 steps per instruction, but you could also add more lines into Table Q2 if required).

Q2c) Could this CPU be a MSP430? State the reason(s) for your answer.

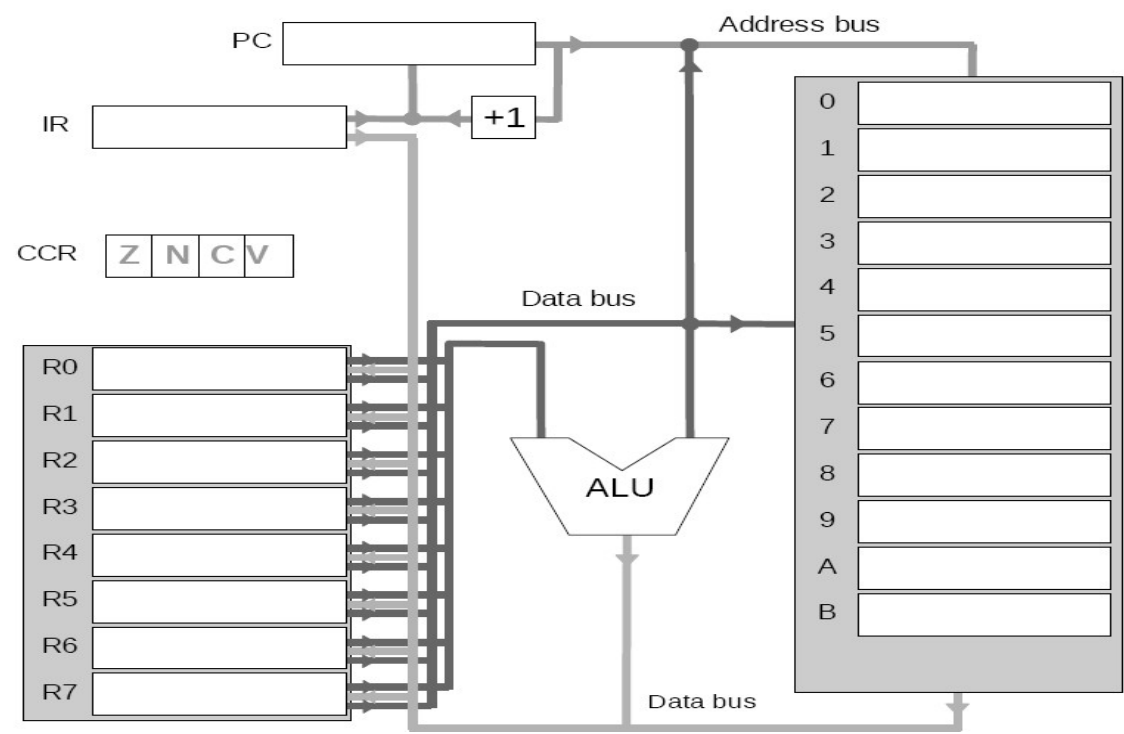


Figure Q2

Instruction	Step	PC	IR	Data Bus	Address Bus	R5	R6
MOV #6, R5	0	0	0	Z (Z means off)	Z (Z means off)	0	0
	1						
	2						
	3						
	4						
	5						
ADD R5,R6	6						
	7						
	8						
	9						
	10						

Table Q2

Q3. Memory Organisation

Q3a) Based on the memory map of a MSP430 microcontroller shown in Figure Q3, draw an equivalent byte-sized memory map.

Q3b) Based on the memory map, give the value of the various data read from the stated memory locations:

- (i) A byte from address **H'2205** (*give answer in binary*)
- (ii) A word starting at address **H'2204** (*give answer in hexadecimal*)
- (iii) A 7 character string from address **H'2202** (*in ASCII character string, e.g. "STRING"*)
(*Note: You may use the hexadecimal to ASCII character set conversion table given*)

Address	Contents
H' 00220A	H' 86B6
H' 002208	H' A531
H' 002206	H' 3130
H' 002204	H' 3154
H' 002202	H' 4349
H' 002200	H' BBE7

LS \ MS	0	1	2	3	4	5	6	7
0	NUL	DLE	SP	0	@	P	'	p
1	SOH	DC1	!	1	A	Q	a	q
2	STX	DC2	"	2	B	R	b	r
3	ETX	DC3	#	3	C	S	c	s
4	EOT	DC4	\$	4	D	T	d	t
5	ENQ	NAK	%	5	E	U	e	u
6	ACK	SYN	&	6	F	V	f	v
7	BEL	ETB	'	7	G	W	g	w
8	BS	CAN	(	8	H	X	h	x
9	HT	EM	)	9	I	Y	i	y
A	LF	SUB	*	:	J	Z	j	z
B	VT	ESC	+	;	K	[	k	{
C	FF	FS	,	<	L	\	l	
D	CR	GS	-	=	M	]	m	}
E	SO	RS	.	>	N	^	n	
F	SI	US	/	?	O	_	o	DEL

ASCII Character Set (7-Bit Code)

Figure Q3